國立臺灣大學 | 生物資源暨農學院
生物機電工程學系
Department of Biomechatronics Engineering, National Taiwan University

BME5934

Introduction to Conda

2025/06/29

Dr. Ping-Han Hsieh

Module, Library, Package and Application

- **Module: a single, self-contained file of code**. Typically contains a small group of related functions and classes designed for a specific purpose.
- **Library:** a collection of **multiple modules** that are all related to a broader domain. A program or application "calls" upon a library to perform a specific function without the need to develop the functions from scratch.
- **Package:** a collection of related **modules, libraries** and other files bundled together **for distribution**.
- **Application**: a complete, **standalone program** designed to perform a specific task for an end-user. Typically provided with a graphical user interface (GUI) or command-line tools.

Package Manager

- Most applications rely on a specific version of shared libraries or packages, which are known as **dependencies**.
- To avoid version conflicts and ensure a consistent environment, developers use a **package manager**. This tool automates the process of installing, updating, and managing a project's dependencies.
- Here, we'll introduce **Conda**, a powerful tool that serves as both a **package manager** and an **environment manager**.
- A **virtual environment** is a self-contained directory that holds a specific version of applications and associated packages or libraries, allowing for **isolating a project's dependencies** from other projects on your system.

Virtual Environment Management (1)

Create Virtual Environment

```
$ conda create -n [ENVIRONMENT-NAME]
```

Example (Terminal)

```
(base) $ conda create -n bme5934
```

Virtual Environment Management (2)

Create Virtual Environment

```
$ conda create -n [ENVIRONMENT-NAME]
```

Switch Virtual Environment

```
$ conda activate [ENVIRONMENT-NAME]
```

Example (Terminal)

```
(base) $ conda create -n bme5934
(base) $ conda activate bme5934
(bme1111) $
```

- Note that once an environment is activated, it remains active and applies to all subsequent commands in that terminal session, regardless of the directory you navigate to.
- Conda manages the required environment variables, install paths, and dependencies for your projects.

Virtual Environment Management (3)

Create Virtual Environment

```
$ conda create -n [ENVIRONMENT-NAME]
```

Switch Virtual Environment

```
$ conda activate [ENVIRONMENT-NAME]
```

Show All Environment

```
$ conda env list
```

Example (Terminal)

```
(base) $ conda create -n bme5934
(base) $ conda activate bme5934
(bme5934) $ conda env list

# conda environments:
#
base             /home/user/miniconda3
bme5934      *  /home/user/miniconda3/envs/bme5934
```

Virtual Environment Management (4)

Create Virtual Environment

```
$ conda create -n [ENVIRONMENT-NAME]
```

Switch Virtual Environment

```
$ conda activate [ENVIRONMENT-NAME]
```

Show All Environment

```
$ conda env list
```

Remove Environment

```
$ conda remove --all -n [ENVIRONMENT-NAME]
```

Example (Terminal)

```
(base) $ conda create -n tmp
(base) $ conda activate tmp
(tmp) $ conda env list

# conda environments:
#
base             /home/user/miniconda3
tmp       * /home/user/miniconda3/envs/tmp

(tmp) $ conda activate base
(base) $ conda remove --all -n tmp
```

- **This will remove all installed libraries, packages, applications in that environment.**
- **Never remove the default (base) environment.**

Channel Management (1)

- Conda package management tool typically manage the available packages in different **channels**.
- Channels act as **package repositories**, storing various versions of packages for different operating system and platform.
- The packages in Conda channels can be maintained by companies, research groups, and the open-source community.
- Check here for the available packages that can be installed using Conda.
- **Not all packages are available in the Conda repositories.**

Channel Management (1)

Check Current Tracked Channel

```
$ conda config --show channels
```

Example (Terminal)

```
(base) $ conda create -n tmp
(tmp) $ conda activate tmp
(tmp) $ conda config --show channels
channels:
  - defaults
```

Channel Management (2)

Check Current Tracked Channel

```
$ conda config --show channels
```

Tracked Specific Channel

```
$ conda config --env --add channels [CHANNEL-NAME]
```

Example (Terminal)

```
(base) $ conda create -n tmp
(tmp) $ conda activate tmp
(tmp) $ conda config --show channels
channels:
  - defaults

(tmp) $ conda config --env --add channels conda-forge
```

- **conda only searches and installs packages in the tracked repositories.**
- **conda-forge is an open-source community-maintained channel which contains many useful packages and libraries.**

Channel Management (3)

Tracked Specific Channel (with Top Priority)

```
$ conda config --env --add channels [CHANNEL-NAME]
```

Tracked Specific Channel (with Lowest Priority)

```
$ conda config --env --append channels [CHANNEL-NAME]
```

Untracked Channel

```
$ conda config --env --remove channels [CHANNEL-NAME]
```

Set Channel Search Rule

```
$ conda config --set channel_priority strict
```

Example (Terminal)

```
(base) $ conda create -n tmp
(tmp) $ conda activate tmp
(tmp) $ conda config --show channels
channels:
  - defaults

(tmp) $ conda config --env --add channels conda-forge
(tmp) $ conda config --env --remove channels conda-forge
```

It is recommended to use "`strict`"; other available options are "`flexible`" or "`disabled`"

Package Management (1)

Check Current Installed Package

```
$ conda list -n [REPOSITORIES]
```

Example (Terminal)

```
(tmp) $ conda list –n base
```

Package Management (2)

Check Current Installed Package

```
$ conda list -n [REPOSITORIES]
```

Search Available Packages

```
$ conda search [PACKAGE-NAME]
```

Example (Terminal)

```
(tmp) $ conda list -n base
(tmp) $ conda search scanpy
...
scanpy         1.11.4    pyhd8ed1ab_0  conda-forge
scanpy         1.11.5    pyhd8ed1ab_0  conda-forge
scanpy         1.12      pyhd8ed1ab_0  conda-forge
scanpy         1.12.1    pyhd8ed1ab_0  conda-forge
```

- **conda only searches and installs packages in the tracked repositories.**

Package Management (3)

Check Current Installed Package

```
$ conda list -n [REPOSITORIES]
```

Search Available Packages

```
$ conda search [PACKAGE-NAME]
```

Install Packages

```
$ conda install [PACKAGE-NAME]
```

Example (Terminal)

```
(tmp) $ conda list -n base
(tmp) $ conda search cmake
...
cmake             4.0.3       h74e3db0_0  conda-forge
cmake             4.1.0       hc85cc9f_0  conda-forge
cmake             4.1.1       hc85cc9f_0  conda-forge

(tmp) $ conda install cmake=4.1.1
```

- **Install can also be used to update the installed packages. You can also check conda update for a more explicit command.**

Package Management (4)

Check Current Installed Package

```
$ conda list -n [REPOSITORIES]
```

Search Available Packages

```
$ conda search [PACKAGE-NAME]
```

Install Packages

```
$ conda install [PACKAGE-NAME]
```

Uninstall Packages

```
$ conda uninstall [PACKAGE-NAME]
```

Example (Terminal)

```
(tmp) $ conda list -n base
(tmp) $ conda search cmake
...
cmake          4.0.3      h74e3db0_0  conda-forge
cmake          4.1.0      hc85cc9f_0  conda-forge
cmake          4.1.1      hc85cc9f_0  conda-forge

(tmp) $ conda install cmake=4.1.1
...

(tmp) $ conda uninstall cmake=4.1.1
...
```

Thank you